AI Assisted Coding

Lab Assignment - 18.1

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Lab 18 – API Integration: Connecting to External Services with Error Handling

Lab Objectives:

- Learn how to integrate Python programs with external REST APIs.
- Understand API request/response handling using requests or similar libraries.
- Implement proper error handling for failed API calls (timeouts, invalid responses, rate limits).
- Practice extracting and displaying meaningful information from API responses.

Task 1 – Weather API Integration

Connect to a Weather API to fetch current weather details for a given city.

Instructions:

- Prompt AI to generate Python code using the requests library.
- Handle errors: invalid city name, API key errors, and network failures.
- Display temperature, humidity, and weather description in a readable format.

Expected Output:

• A working Python script that fetches and displays weather data with proper error handling.

Copilot Suggested Code :

task_1.py > ...

```python
# Connect to a Weather API to fetch current weather details for a given city.

import requests

def get_weather(city_name):
    # Step 1: Get the latitude and longitude of the city using the Geocoding API
    geocoding_url = f"https://geocoding-api.open-meteo.com/v1/search?name={city_name}"
    geocoding_response = requests.get(geocoding_url)

    if geocoding_response.status_code != 200:
        print("Error fetching geocoding data.")
        return

    geocoding_data = geocoding_response.json()

    if not geocoding_data['results']:
        print("City not found.")
        return

    latitude = geocoding_data['results'][0]['latitude']
    longitude = geocoding_data['results'][0]['longitude']

    # Step 2: Get the current weather details using the Open-Meteo API
    weather_url = f"https://api.open-meteo.com/v1/forecast?latitude={latitude}&longitude={longitude}&current_weather=true"
    weather_response = requests.get(weather_url)

    if weather_response.status_code != 200:
        print("Error fetching weather data.")
        return

    weather_data = weather_response.json()

    current_weather = weather_data.get('current_weather', {})

    if not current_weather:
        print("Current weather data not available.")
        return

    temperature = current_weather.get('temperature')
    wind_speed = current_weather.get('wind_speed')

    print(f"Current weather in {city_name}:")
    print(f"Temperature: {temperature}°C")
    print(f"Wind Speed: {wind_speed} km/h")

# Example usage
city = input("Enter the name of the city: ")
get_weather(city)
```

Output :

```
PS C:\B.Tech\3-2\AI_Assisted_Coding\lab_assignment-18> & "C:\Program Files\Python314\python.exe" c:/B.Tech/3-2/AI_Assisted_Coding/lab_assignment-18/t
ask_1.py
Enter the name of the city: warangal
Current weather in warangal:
Temperature: 30.1°C
Wind Speed: None km/h
PS C:\B.Tech\3-2\AI_Assisted_Coding\lab_assignment-18>
```

Task 2 – Currency Exchange Rates

Integrate a Currency Exchange API to convert INR to USD, EUR, and GBP.

Instructions:

- Use AI to generate API request code.
- Validate user input for amount and currency codes.
- Implement error handling for invalid responses and API downtime.
- Show conversion results clearly in tabular format.

Expected Output:

- A script that converts INR to multiple currencies with graceful error handling.

Copilot Suggested Code :

task_2.py > ...

```python
# Integrate a Currency Exchange API to convert INR to USD, EUR, and GBP.

import requests
import pandas as pd

def convert_currency(amount_inr):
    # Validate user input
    if amount_inr <= 0:
        print("Please enter a valid amount greater than 0.")
        return

    # Step 1: Get the exchange rates for INR to USD, EUR, and GBP
    exchange_url = "https://api.exchangerate-api.com/v4/latest/INR"
    try:
        exchange_response = requests.get(exchange_url)
        exchange_response.raise_for_status()  # Check for HTTP errors
    except requests.exceptions.RequestException as e:
        print(f"Error fetching exchange rates: {e}")
        return

    exchange_data = exchange_response.json()

    if 'rates' not in exchange_data:
        print("Exchange rate data not available.")
        return

    rates = exchange_data['rates']

    # Get the conversion rates for USD, EUR, and GBP
    usd_rate = rates.get('USD')
    eur_rate = rates.get('EUR')
    gbp_rate = rates.get('GBP')

    if not all([usd_rate, eur_rate, gbp_rate]):
        print("One or more currency rates are not available.")
        return

    # Step 2: Convert the amount from INR to USD, EUR, and GBP
    converted_amounts = {
        'Currency': ['USD', 'EUR', 'GBP'],
        'Amount': [amount_inr * usd_rate, amount_inr * eur_rate, amount_inr * gbp_rate]
    }

    # Display the results in tabular format
    df = pd.DataFrame(converted_amounts)
    print(df)

# Example usage
try:
    amount = float(input("Enter the amount in INR: "))
    convert_currency(amount)

except ValueError:
    print("Invalid input. Please enter a numeric value for the amount.")
```

Output :

```
PS C:\B.Tech\3-2\AI_Assisted_Coding\lab_assignment-18> & "C:\Program Files\Python314\python.exe" c:/B.Tech/3-2/AI_Assisted_Coding/lab_assignment-18/t
ask_2.py
Enter the amount in INR: 9999999
  Currency        Amount
0      USD  105999.98940
1      EUR   91899.99081
2      GBP   79599.99204
PS C:\B.Tech\3-2\AI_Assisted_Coding\lab_assignment-18>
```

Task 3 – GitHub Repository Info Fetcher

Connect to the GitHub API to fetch details of a repository (e.g., stars, forks, issues).

Instructions:

- Prompt AI to write a Python function to send GET requests to GitHub API.
- Handle rate limit errors, 404 (repo not found), and invalid input.
- Display repository name, description, stars, forks, and open issues.

Expected Output:

- A Python program that retrieves and displays GitHub repository information with robust error handling.

Copilot Suggested Code :

task_3.py > ...

```python
# Connect to the GitHub API to fetch details of a repository (e.g., stars, forks, issues).

import requests

def fetch_github_repo_info(owner, repo):
    # Validate user input
    if not owner or not repo:
        print("Please provide both the repository owner and name.")
        return

    # GitHub API URL for repository information
    api_url = f"https://api.github.com/repos/{owner}/{repo}"

    try:
        response = requests.get(api_url)
        response.raise_for_status()  # Check for HTTP errors

        repo_data = response.json()

        # Display repository information
        print(f"Repository Name: {repo_data.get('name')}")
        print(f"Description: {repo_data.get('description')}")
        print(f"Stars: {repo_data.get('stargazers_count')}")
        print(f"Forks: {repo_data.get('forks_count')}")
        print(f"Open Issues: {repo_data.get('open_issues_count')}")

    except requests.exceptions.HTTPError as http_err:
        if response.status_code == 404:
            print("Repository not found. Please check the owner and repository name.")
        elif response.status_code == 403:
            print("Rate limit exceeded. Please try again later.")
        else:
            print(f"HTTP error occurred: {http_err}")
    except requests.exceptions.RequestException as err:
        print(f"Error fetching repository information: {err}")

# Example usage
owner = input("Enter the GitHub repository owner: ")
repo = input("Enter the GitHub repository name: ")
fetch_github_repo_info(owner, repo)
```

Output :

```
PS C:\B.Tech\3-2\AI_Assisted_Coding\lab_assignment-18> & "C:\Program Files\Python314\python.exe" c:/B.Tech/3-2/AI_Assisted_Coding/lab_assignment-18/task_3.py
Enter the GitHub repository owner: ItzAquarius
Enter the GitHub repository name: mern-ci-project
Repository Name: mern-ci-project
Description: None
Stars: 0
Forks: 0
Open Issues: 0
PS C:\B.Tech\3-2\AI_Assisted_Coding\lab_assignment-18>
```

Task 4 – Real-Time Application: News Headlines Aggregator Scenario: Build a News Aggregator using a free News API.

Requirements:

- Fetch top 5 headlines for a given category (sports, technology, health).
- Handle errors like invalid category, missing API key, and request failures.
- Display headlines in a user-friendly numbered list format.
- Include a retry mechanism if the first request fails.

Expected Output:

- A working Python script that retrieves and displays news headlines with proper error handling.

Copilot Suggested Code :

task_4.py > ...

```python
# Build a News Aggregator using a free News API.

import requests
import time
API_KEY = 'your_api_key_here'  # Replace with your actual API key
BASE_URL = 'https://newsapi.org/v2/top-headlines'

def fetch_headlines(category):
    params = {
        'apiKey': API_KEY,
        'category': category,
        'country': 'us',
        'pageSize': 5
    }
    try:
        response = requests.get(BASE_URL, params=params)
        response.raise_for_status()  # Check for HTTP errors
        data = response.json()
        if data['status'] != 'ok':
            print(f"Error fetching news: {data.get('message', 'Unknown error')}")
            return []
        return data['articles']
    except requests.exceptions.RequestException as e:
        print(f"Request failed: {e}")
        return []

def display_headlines(headlines):
    if not headlines:
        print("No headlines to display.")
        return
    print("\nTop 5 Headlines:")
    for idx, article in enumerate(headlines, start=1):
        print(f"{idx}. {article['title']}")

def main():
    category = input("Enter news category (sports, technology, health): ").strip().lower()
    if category not in ['sports', 'technology', 'health']:
        print("Invalid category. Please choose from sports, technology, or health.")
        return

    headlines = fetch_headlines(category)
    if not headlines:
        print("Retrying to fetch headlines...")
        time.sleep(2)  # Wait before retrying
        headlines = fetch_headlines(category)

    display_headlines(headlines)

if __name__ == "__main__":
    main()
```

Output :

```
PS C:\B.Tech\3-2\AI_Assisted_Coding\lab_assignment-18> & "C:\Program Files\Python314\python.exe" c:/B.Tech/3-2/AI_Assisted_Coding/lab_assignment-18/t
ask_4.py
Enter news category (sports, technology, health): sports
Request failed: 401 Client Error: Unauthorized for url: https://newsapi.org/v2/top-headlines?apiKey=your_api_key_here&category=sports&country=us&page
Size=5
Retrying to fetch headlines...
Request failed: 401 Client Error: Unauthorized for url: https://newsapi.org/v2/top-headlines?apiKey=your_api_key_here&category=sports&country=us&page
Size=5
No headlines to display.
PS C:\B.Tech\3-2\AI_Assisted_Coding\lab_assignment-18>
```